E-Commerce Customer Drop-off & Conversion Analysis

End-to-end Data Analyst Portfolio Project

Tools	SQL Server • Power BI • Python for data preparation (include only if used in the final workflow)
Analysis period	October 1, 2019 event data
Dataset	Multi-category e-commerce behavior events
Report purpose	Explore session conversion, drop-off points, category performance, and price-band patterns.

Executive summary

This project analyzes e-commerce browsing and transaction events at session level to understand where shopping journeys lose momentum. SQL Server was used to validate the event data and build session-level funnel tables, while Power BI was used to present overall funnel performance, category-level conversion, and price-band patterns.

The clearest signal in the sequential funnel is the transition from product view to cart addition: approximately 4.31% of view sessions progressed to a cart, equivalent to about 95.69% not reaching the cart stage. Among sessions that added a cart, approximately 56.77% also recorded a purchase in the cart-related analysis. These results identify areas for further investigation; they do not establish the causes of customer behavior.

Important scope note: the prepared data covers October 1, 2019 from 00:00 through 22:43, not the entire month. All conclusions should be interpreted within that limited time window.

1. Business Problem & Objectives

Business problem. The e-commerce business lacks a clear view of where sessions drop out of the shopping journey and how conversion varies by product category, brand, price band, and browsing behavior.

Primary objective. Quantify session-level progression through view, cart, and purchase stages and identify the largest drop-off points.

Supporting objectives.

- Measure view, cart, and purchase sessions using session-level logic rather than raw event counts.
- Compare conversion rates across categories and price bands.
- Explore associations between purchasing and browsing behaviors.
- Translate observed patterns into practical questions and improvement opportunities.

Stakeholders

E-commerce/product managers, marketing teams, category managers, sales leaders, user-experience teams, and data/analytics teams.

2. Dataset & Data Preparation

The project uses the Kaggle dataset “eCommerce behavior data from multi-category store.” The prepared file contains 499,712 rows and 16 columns, including event timestamp, event type, product/category identifiers, brand, price, user ID, session ID, event date/hour, day of week, price band, and time band.

Validation item	Result
Prepared event records	499,712
Unique users	89,124
Unique sessions	117,455
View events	481,790
Cart events	8,165
Purchase events	9,757
Duplicate rows reported	0
Null category_code values	158,366
Null brand values	71,843

Preparation and validation work: checked record counts and duplicates, reviewed missing category and brand values, derived date/time and price-band attributes, and created session-level tables in SQL Server for overall, category, brand, price-band, and hour analysis.

Data quality consideration: category and brand analyses exclude or group records where those fields are missing, so their coverage differs from the overall session funnel.

3. Overall Funnel Analysis

Sequential stage	Sessions	Interpretation
View	117,406	Sessions with a view
Cart	5,066	View sessions that also added to cart
Purchase	2,875	View sessions that also added to cart and purchased

Key rates

Metric	Approx. result	Definition
View → Cart	4.31%	Sequential cart sessions ÷ view sessions
View → Cart drop-off	95.69%	100% – View → Cart conversion
Cart → Purchase	56.77%	Sessions with cart and purchase ÷ sessions with cart (cart-added analysis)
View → Purchase (sequential)	2.45%	Sequential purchase sessions ÷ view sessions

Interpretation. The most prominent leakage occurs between product views and cart additions. This suggests the product discovery and product-detail journey merits investigation. The data alone cannot tell whether the issue is product relevance, price, delivery cost, product information, or interface usability.

Metric distinction. The Dashboard 1 purchase KPI counts all sessions with any purchase (approximately 8,256, displayed as 8K). The sequential funnel's purchase stage counts only sessions with view + cart + purchase (2,875, displayed as 3K). Both can be correct because they answer different questions.

Potential investigation areas

- Review product-page engagement and Add to Cart button visibility.
- Check whether shipping costs, delivery estimates, and return information are clear before cart addition.
- Analyze checkout friction, payment availability, and unexpected costs for sessions that have added items to cart.

4. Category Performance

The category dashboard compares session volume and conversion across product categories, including top categories by view-to-purchase conversion, traffic versus conversion, funnel conversion, cart volume, abandonment, and purchase sessions.

Example category	View sessions	Cart sessions	Purchase sessions
electronics.smartphone	39,204	3,606	2,232

This category has substantial activity in the analyzed data. Its performance should be assessed alongside conversion rate, not only volume. Category-level values may not sum to overall funnel values because a session can interact with multiple categories and records with missing category codes are not represented in the same way.

5. Price-Band Analysis

Price band	View sessions	Cart sessions	Purchase sessions	View→ Cart	Cart→Purchase	View→Purchase
100–250	46,724	1,920	1,120	4.11%	58.33%	2.40%
250–500	33,504	1,215	746	3.63%	61.40%	2.23%
500–1,000	23,621	841	513	3.56%	61.00%	2.17%
1,000+	12,805	419	237	3.27%	56.56%	1.85%
50–100	27,277	330	160	1.21%	48.48%	0.59%
0–50	31,283	481	177	1.54%	36.80%	0.57%

Observations. The 100–250 band has the highest view volume among the listed price bands and the highest view-to-purchase rate in this summary. The 0–50 and 50–100 bands show lower view-to-purchase rates. These differences may reflect product mix, intent, data coverage, or other factors; price alone should not be assumed to cause the pattern.

Important validation note. This table reflects the previously shared SQL summary. Ensure the Power BI price-band measures use the same sequential definitions before treating every dashboard figure as directly comparable to this table.

6. Browsing Behavior Analysis

Views per session	Sessions	Purchase sessions	View-group purchase rate
1 view	44,034	1,462	3.32%
2–3 views	34,712	3,225	9.29%
4–5 views	14,696	1,371	9.33%
6+ views	24,013	2,198	9.15%

Sessions with two or more views show higher purchase conversion than single-view sessions in the analyzed data, but conversion does not keep increasing as view count rises. This is an association, not evidence that additional views cause a purchase.

Cart presence comparison

Sessions with a cart recorded 2,882 purchase sessions out of 5,077 sessions (56.77%). Sessions without a cart recorded 5,374 purchases out of 112,378 sessions (4.78%). This compares any purchase within each group; it should not be described as a strictly sequential funnel rate.

Product variety

Purchase conversion was 7.34% for sessions interacting with one product, 7.25% for 2–3 products, 6.27% for 4–5 products, and 5.82% for 6+ products. Higher product variety was associated with lower conversion in this sample. Further investigation is needed to determine whether this reflects comparison shopping, product availability, or differences in customer intent.

7. Recommendations & Next Steps

Prioritize the view-to-cart stage: Review product detail pages, product relevance, price clarity, delivery estimates, stock status, and Add to Cart visibility.

Reduce uncertainty before checkout: Make delivery fees, delivery dates, returns, warranties, and payment methods easy to find.

Segment conversion monitoring: Track view-to-cart and cart-to-purchase by category, price band, brand, device, and time when those fields are available.

Investigate low-conversion segments: Review the product mix and user intent in the 0–50 and 50–100 bands before making pricing decisions.

Support product evaluation: Test whether comparison tools, specifications, reviews, and clearer product information help customers decide.

Validate with a longer time window: Repeat the analysis across additional dates and compare results by weekday, hour, and campaign to check stability.

8. Methodology, Definitions & Limitations

Analysis grain

Session-level funnel metrics use unique session identifiers and flags for whether a session contains a view, cart, or purchase. Sequential metrics additionally require each prior stage to be present. Category and price-band analyses are calculated within their respective grouping tables and should be interpreted at that grain.

Core definitions

Metric	Definition
View → Cart %	Sessions with View and Cart ÷ sessions with View
Cart → Purchase %	Sessions with Cart and Purchase ÷ sessions with Cart
Sequential View → Purchase %	Sessions with View, Cart, and Purchase ÷ sessions with View
Drop-off %	100% – the corresponding conversion rate

Limitations

- The prepared event data covers October 1, 2019 only, from 00:00 to 22:43; it is not a full-month dataset.
- The analysis is observational and descriptive; it cannot establish causation.
- Missing category and brand values reduce coverage for those specific analyses.
- A session can include multiple products or categories, so category totals may overlap and differ from overall totals.
- Purchase events can occur in sessions without a recorded cart; therefore, all-purchase session counts and sequential funnel purchase counts differ.
- Results depend on the event definitions, sessionization, and price-band assignment used during preparation.

Conclusion

The analysis identifies product view to cart addition as the largest observed funnel drop-off and highlights differences in conversion by category, price band, and browsing behavior. The dashboards provide a practical monitoring layer for e-commerce stakeholders. The next step is to validate these patterns over a broader time window and test targeted improvements before attributing business impact.

Dataset source

Kaggle — “eCommerce behavior data from multi-category store” by Michael Kechinov / Open CDP / REES46. Dataset page: https://www.kaggle.com/datasets/mkechinov/ecommerce-behavior-data-from-multi-category-store